MA.6.NSO.1.2

Benchmark

MA.6.NSO.1.2 **Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.**

Example: Jasmine is on a cruise and is going on a scuba diving excursion. Her elevations of 10 feet above sea level and 8 feet below sea level can be compared on a number line, where 0 represents sea level.

Benchmark Clarifications:
Clarification 1: Instruction includes vertical and horizontal number lines, context referring to distances, temperatures and finances and using informal verbal comparisons, such as, lower, warmer or more in debt.
Clarification 2: Within this benchmark, the expectation is to compare positive and negative rational numbers when given in the same form.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.4.1, MA.6.NSO.4.2
- MA.6.AR.1.1, MA.6.AR.1.2
- MA.6.GR.1.1, MA.6.GR.1.2, MA.6.GR.1.3
- MA.7.NSO.2.1

Terms from the K-12 Glossary

- Integers
- Number Line
- Rational Number
- Whole Number

Vertical Alignment

Previous Benchmarks

- This is the first introduction to the concept of numbers having opposites.

Next Benchmarks

- MA.7.NSO.1

Purpose and Instructional Strategies

In previous courses, students plotted, ordered and compared positive numbers. In grade 6 accelerated, students plot, order and compare on both sides of zero on the number line with all forms of rational numbers. This benchmark focuses on the comparison of quantities with opposite directions (one positive and one negative) and understanding what the zero represents within the provided context. Students will also use the concept of opposites when solving problems involving order of operations and absolute value. In future courses, students will plot, order and compare positive and negative rational numbers.

- A strong foundation in this skill can help students more efficiently solve problems involving absolute value, finding the distance between points on a coordinate plane, combining rational numbers as well as assessing the reasonableness of solutions.

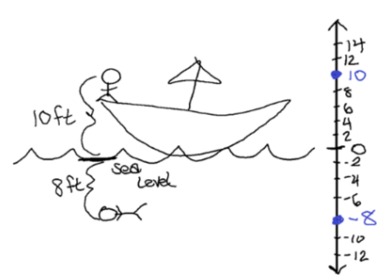
- Within this benchmark, it is the expectation that students make informal verbal comparisons of rational numbers using "is greater than" or "is less than" or another comparison within context.
  - For example, when comparing the temperature in Chipley, Florida, which is 34°F, to Minneapolis, Minnesota, which is -4°F, one could say that it is warmer in Florida than Minnesota or that it is colder in Minnesota than Florida.
- Students can describe the relative relationships between quantities and include the distance between the values in their description, as this is a way of connecting and reinforcing the MA.6.GR.1 standard *(MTR.2.1, MTR.7.1)*.
  - For example, if a shark is 7.5 meters below sea level and a cliff diver is standing on a cliff at a point that is 22 meters from the surface of the water, the shark is 29.5 meters below the cliff diver. The shark's position can also be described as -7.5 meters from the surface of the water if the surface of the water represents 0.

Common Misconceptions or Errors

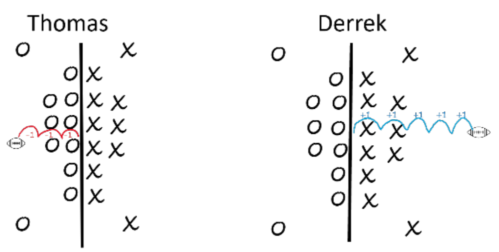
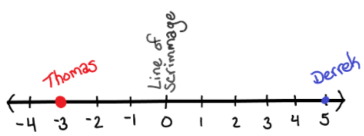
- Students may incorrectly assign double negatives when describing relationships that are below zero.
  - For example, they may say the temperature is -3 degrees below zero when they mean it is -3 degrees from zero or it is 3 degrees below zero. In this same context, it is important for students to understand that identifying a value as negative indicates the position as being lower or colder than the neutral position of 0.
- Instruction provides many opportunities for students to share verbal descriptions (in written and oral forms) to better develop this skill (*MTR.4.1, MTR.7.1*).
- Students may experience difficulty when trying to create a number line and determining the meaning of 0 within the context. Instruction showcases drawing a picture of the context first and then connecting it to the building of a number line.

Strategies to Support Tiered Instruction

- Instruction includes the use of a physical or digital number line to plot various values and to use their location on the number line to assist in comparing them. Instruction focuses on the generalization that the farther to the left a value is on the number line, the lesser the number.
- Teacher assists students with representing rational values in a real-world context, by first drawing a visual representation of the situation and labeling key features of the picture such as the "neutral" point of the visual (sea level, zero degrees, ground level, etc.), direction (above or below sea level, forward or backwards movement, etc.) and magnitude (the absolute value of the given quantity) of the given situation. Once a visual is drawn, the teacher or students create a corresponding horizontal or vertical number line to match the picture and the key features and then use the number line to write the correct rational number, paying close attention to the location of the point relative to zero.
  - For example, Jasmine is on a cruise and is going on a scuba diving excursion. Her elevations of 10 feet above sea level and 8 feet below sea level can be compared on a number line, where 0 represents sea level.



- Teacher provides opportunities for students to represent verbal descriptions of situations pictorially and on a number line. Students then restate the same situation using different vocabulary and check the reasonableness of the new situations by making connections back to the visual representation, number line, and meaning of zero in the context.
    - For example, on Saturday, Thomas's football team lost 3 yards in their first drive while on Thursday, Derrek's football team gained 5 yards in their first drive. The first drives for each team can be compared on a number line, where 0 represents the line of scrimmage for each play.

Picture	**Meaning of Zero**
	The start of each first drive.
Number Line	**Additional Description**
	The first drive of Thomas's football team can be represented by -3 and the first drive of Derrek's team can be represented by 5.

- Instruction includes providing students with opportunities to practice placing rational number cards on a provided number line. While placing the cards, students should share their thinking out loud while the teacher listens for reasonable estimations and assumptions for correct placement.
    - For example, if a student is placing a number card with $-4\frac{3}{5}$ on the number line, the student needs to know that $\frac{3}{5}$ is more than half, so to place the card, the student should be between -4 and -5, with the point closer to the -5. As more cards are added to the number line, the students may adjust the placement of other cards, if necessary, for more accurate estimates.

- Teacher and students co-create a list of common terms from contextual situations that may be used to describe positive or negative values.
  - Examples of this could include:

Positive	Negative
Deposit	Withdrawal
Gain	Loss
Above	Below

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.7.1)

On Thursday, Calvin borrowed \$46.68 from his mom to purchase a new video game. He mowed yards on the weekend to earn money to pay his mom back. After paying his mom back, Calvin had \$12.32 left on Monday. What is the meaning of 0 in this situation? Compare the amounts of money that Calvin had on Thursday and Monday. Explain your reasoning including a number line in your justification.

Instructional Task 2 (MTR.5.1)

Part A. Given the inequality $7.2 > -4.5$, describe how the numbers would be positioned relative to each other on a number line.

Part B. What role does 0 play in this context?

Part C. Using this reasoning, describe how any number x and any number y could be positioned relative to each other on a number line and the role played by 0.

Instructional Items

Instructional Item 1

New Orleans, Louisiana has an altitude of about $-6\frac{1}{2}$ feet and Miami, Florida has an altitude of about $6\frac{3}{5}$ feet. Compare the two altitudes on a vertical number line.

Instructional Item 1

Students from Hope Middle School are traveling to Washington D.C. for their grade 8 field trip. When they left Florida, the temperature was 67°F. At their arrival in their hotel in Washington D.C., the temperature was -15°F, which is the coldest day on record in Washington D.C. On their second day in Washington D.C., the temperature was 24°F. Compare the three temperatures on a number line and explain where zero would be on the number line, as well as its relationship to the temperatures.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*